Vishakha Ramani

Research Staff Member, IBM Thomas J. Watson Research Center
Yorktown Heights, New York, USA

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100+ citations across peer-reviewed publications · 11 publications · IEEE Member

Professional Profile

Research Staff Member at IBM Thomas J. Watson Research Center—one of the world's largest and most cited industrial research laboratories—working at the intersection of rigorous performance modeling and production-scale LLM inference systems. Co-architect of the Workload Variant Autoscaler (WVA), a core autoscaling component of llm-d, an open-source Kubernetes-native LLM inference system with 3,000+ GitHub stars adopted by the broader cloud-native AI community. Author of peer-reviewed publications at premier venues including IEEE INFOCOM, IEEE ISIT, and WiOpt, with work accumulating 105+ citations. Selected by leading IEEE journals and conferences as an expert peer reviewer, a designation reserved for recognized specialists in the field.

Education

Rutgers University–New Brunswick 2020–2024
Doctor of Philosophy, Electrical and Computer Engineering (GPA: 3.9/4.0)
Dissertation: *Storing, Retrieving, and Processing Updates: A Timeliness Perspective*
Advisor: Prof. Roy D. Yates
WINLAB (NSF-funded Wireless Information Networking Laboratory)

Rutgers University 2017–2019
Master of Science, Electrical and Computer Engineering (GPA: 3.9/4.0)
Thesis: *I-Mac: An ICN Based Radio Access Network Architecture*
Advisor: Prof. Dipankar Raychaudhuri

Rajiv Gandhi Prodyogiki Vishwavidyalaya 2011–2016
Dual Degree: Bachelor of Technology (BTech) and Master of Technology (MTech), Electronics and Communication Engineering

Professional Experience

Research Staff Member October 2024–Present
IBM Thomas J. Watson Research Center, Yorktown Heights, NY
IBM Research is one of the world's largest and most cited industrial research organizations, with contributions foundational to modern computing.

- **Co-architect of the Workload Variant Autoscaler (WVA)** for *llm-d*, an open-source Kubernetes-native LLM inference system (GitHub). WVA is documented as a core component in the official llm-d architecture (llm-d.ai/docs/architecture), a production system with 3,000+ GitHub stars and regular versioned releases adopted across the cloud-native AI community. Produced the first end-to-end system design and led the autoscaling effort in close collaboration with the broader llm-d community.
- **Designed and implemented the queueing model analyzer**, the analytical core of WVA. The analyzer builds a hardware-aware performance model of LLM inference to predict key serving metrics (time-to-first-token and inter-token latency) and compute the maximum sustainable re-

quest rate per replica under SLO constraints. Model parameters are learned online via a Kalman filter, enabling continuous adaptation to workload shifts without offline profiling.

- Co-authored the WVA system paper: A. Malvankar et al., "WVA: A Global Optimization Control Plane for llm-d," to appear in *Proc. IEEE International Conference on Cloud Computing (CLOUD)*, 2026 (arXiv:2603.09730).
- Contributed original analysis to T. Abdelzaher et al., "The Bottlenecks of AI," *Real-Time Systems*, vol. 61, 2025 [**12 citations**], identifying and framing the open research problems in real-time cloud infrastructure for LLM inference: how to scale inference serving systems to many concurrent models and users while satisfying latency SLOs, managing GPU memory efficiently, and orchestrating heterogeneous accelerator pools.
- Developed a rigorous analytical framework for prefill/decode (P/D) disaggregation in LLM inference, deriving a dynamic scheduling threshold via Lyapunov optimization on a tandem queueing model with provable throughput and latency guarantees (manuscript in preparation).
- **Analytical guarantees for agent-discovered algorithms.** Applied the AI-Driven Research for Systems (ADRS) methodology of Liu et al. (arXiv:2510.06189) to scheduling and autoscaling problems in llm-d. Contributed analytical performance models that supply provable guarantees for algorithms discovered by the agentic search loop, complementing the empirical evaluation produced by the search itself. Related open-source artifacts produced by the team: BLIS (the simulation substrate) and Agentic Strategy Evolution (the search loop).

Graduate Research Assistant September 2018–October 2024
Rutgers University–New Brunswick, New Brunswick, NJ
PhD Research — Age of Information (AoI) in Computing Systems

- **AoI in shared memory and synchronization:** Analyzed the impact of synchronization primitive choice (lock-based reader-writer locks vs. lock-free Read-Copy-Update) on information freshness in shared-memory status-updating systems. Proved that the optimal choice depends on the update rate regime, with lock-based primitives favoring high-rate and lock-free favoring low-rate workloads. Characterized fundamental age-memory trade-offs in RCU systems (IEEE INFOCOM 2023; IEEE INFOCOM 2024 Workshop).
- **Optimal memory access policies:** Formulated an optimal memory sampling problem in status-updating systems and established that the optimal policy is a stationary, deterministic threshold-type policy, deriving closed-form AoI expressions under this policy (IEEE ISIT 2024).
- **Multi-source and multi-step update processing:** Developed analytical models for AoI in publish-subscribe systems with multiple update sources; showed that a lazy computation policy can reduce average AoI at the monitor (WiOpt 2023). Extended the framework to multi-step sequential processing pipelines, identifying wasted computation that arises when processing effort does not reduce age, and optimizing service rates under power constraints (Asilomar 2024).
- **Experimental timeliness in mobile networks:** Conducted experimental studies of timely packet routing using DPDK-based high-speed testbeds, demonstrating the sensitivity of AoI to offered load and concurrency constructs in real network deployments (IEEE INFOCOM 2023 Workshop).

MS Research — ICN-based Wireless Access Networks

- Designed a novel Information-Centric Networking (ICN) based Radio Access Network (RAN) protocol for LTE, enabling dynamic and efficient wireless multicast via a new cross-layer architecture integrating ICN identifiers and semantics with LTE RAN. Verified the design with proof-of-concept experiments using NS-3 and SUMO simulations.

- Designed a scalable ICN-based MAC scheduling algorithm using convex optimization for joint resource allocation across unicast and multicast user groups.

Research Intern May–August 2023
IBM Thomas J. Watson Research Center, Yorktown Heights, NY

- Conducted performance analysis of the Multi-Cluster App Dispatcher (MCAD) using Kubernetes Without Kubelet (KWOK) to identify and eliminate scheduling bottlenecks in cloud-native foundational model workloads.
- Presented findings at **KubeCon + CloudNativeCon North America 2023** (Chicago, IL), the world's largest cloud-native computing conference, drawing over 13,000 registered attendees from industry and academia.

Research Intern May–August 2022
IBM Thomas J. Watson Research Center, Yorktown Heights, NY

- Designed and implemented a middleware systems stack for scalable distributed training of foundational models on OpenShift as part of **Project Codeflare**, an open-source IBM initiative for cloud-native ML workload orchestration.
- Contributed open-source code to the **TorchX** project, a PyTorch-ecosystem tool for defining and running ML pipelines.
- Conducted research and benchmarking on multi-tenant cluster schedulers to evaluate scheduling efficiency under concurrent foundational model training workloads.

Assistant System Engineer February–July 2017
Tata Consultancy Services, Mumbai, India

- Business intelligence and performance management; developed BI solutions using IBM Cognos.

Publications

Journal Articles

1. T. Abdelzaher, Y. Hu, D. Kara, T. Kimura, A. Misra, **V. Ramani**, O. Tardieu, et al., "The Bottlenecks of AI: Challenges for Embedded and Real-Time Research in a Data-Centric Age," *Real-Time Systems*, vol. 61, no. 2, pp. 185–236, 2025. [**12 citations**] *(Contributed original analysis of real-time cloud challenges for large-scale LLM inference serving, including SLO-driven autoscaling, prefill/decode disaggregation, distributed KV-cache management, and multi-tenant inference orchestration.)*
2. **V. Ramani** and S. K. Sharma, "Cognitive Radios: A Survey on Spectrum Sensing, Security and Spectrum Handoff," *China Communications*, vol. 14, no. 11, pp. 185–208, 2017. [**84 citations**]

Conference Papers

1. A. Malvankar, L. Villard, M. Abdi, E. Shindin, B. Dumba, **V. Ramani**, A. Tantawi, et al., "WVA: A Global Optimization Control Plane for llm-d," to appear in *Proc. IEEE International Conference on Cloud Computing (CLOUD)*, 2026 (arXiv:2603.09730).
2. **V. Ramani**, I. Seskar, and R. D. Yates, "Timely and Energy-Efficient Multi-Step Update Processing," in *Proc. 58th Asilomar Conference on Signals, Systems, and Computers*, pp. 116–120, 2024.

3. **V. Ramani**, I. Seskar, and R. D. Yates, "Efficient and Timely Memory Access," in *Proc. IEEE International Symposium on Information Theory (ISIT)*, pp. 1155–1160, 2024.
4. **V. Ramani**, I. Seskar, and R. D. Yates, "Timely Processing of Updates from Multiple Sources," in *Proc. 21st International Symposium on Modeling and Optimization in Mobile, Ad Hoc, and Wireless Networks (WiOpt)*, 2023.
5. **V. Ramani**, J. Chen, and R. D. Yates, "Lock-Based or Lock-Less: Which is Fresh?" in *Proc. IEEE International Conference on Computer Communications (INFOCOM)*, pp. 1–10, 2023. [**3 citations**]

Workshop Papers

1. **V. Ramani**, J. Chen, and R. D. Yates, "Age-Memory Trade-Off in Read-Copy-Update," in *IEEE INFOCOM 2024 Workshops*, 2024.
2. **V. Ramani**, J. Chen, and R. D. Yates, "Timely Mobile Routing: An Experimental Study," in *IEEE INFOCOM 2023 Workshops*, 2023. [**3 citations**]

Working Papers

1. **V. Ramani** and A. N. Tantawi, "An Approximate Parameterized Queueing Model of LLM Inference Serving," *Manuscript in preparation.* Develops a tractable Markovian queueing model for LLM inference serving; derives closed-form mean ITL and numerically evaluates TTFT; validated against real GPU measurements.
2. **V. Ramani**, "When to Disaggregate: Optimal Online Scheduling for Prefill-Decode Separated LLM Inference," *Manuscript in preparation.*

Dissertations

1. **V. Ramani**, "Storing, Retrieving, and Processing Updates: A Timeliness Perspective," *PhD Dissertation*, Rutgers University, 2024.
2. **V. Ramani**, "I-Mac: An ICN Based Radio Access Network Architecture," *MS Thesis*, Rutgers University, 2020.

Invited Talks and Presentations

- "Performance Analysis of Multi-Cluster App Dispatcher (MCAD)," **KubeCon + CloudNativeCon North America 2023**, Chicago, IL, November 2023. *KubeCon is the world's largest cloud-native computing conference, with over 8,000 registered attendees; selected via competitive proposal review by the Cloud Native Computing Foundation (CNCF).*
- "Using Age of Information (AoI) Models to Enable Low Latency Services," **Invited Talk, Fall 2022 Research Review, WINLAB, Rutgers University**, Fall 2022. *WINLAB is an NSF-funded research center recognized internationally for wireless and mobile networking research.*

Professional Service

Program Committees

- **Program Committee Member**, AAAI/ACM Conference on AI, Ethics, and Society (AIES) 2026, Malmö, Sweden, October 2026. *AIES is a joint conference of the Association for the Advancement of Artificial Intelligence (AAAI) and the Association for Computing Machinery (ACM); PC members are selected by the Program Co-Chairs to evaluate submissions on behalf of the research community.*

Journal and Conference Reviewing

Selected by the following premier venues as an expert reviewer:

- *IEEE Journal on Selected Areas in Communications* (JSAC) — one of the highest-impact journals in communications engineering
- *IEEE Transactions on Wireless Communications* (TWC) — flagship journal of the IEEE Communications Society
- *IEEE Transactions on Communications* (TCOM)
- *IEEE International Symposium on Information Theory* (ISIT) — premier venue for information theory; highly selective
- *IEEE Internet of Things Journal*
- *Ad Hoc Networks* (Elsevier)
- *BalkanCom*

Invention Disclosures

- A. Malvankar, A. Tantawi, **V. Ramani**, T. Eilam, M. Abdi, "System and Method for Efficient SLO-Aware Autoscaling of AI Inference Workloads," IBM Invention Disclosure P202503087 (Record ID 99694127). Rated for defensive publication; article in preparation at IP.com, 2025.

Honors, Awards, and Memberships

- **Competitive Travel Grant**, Cloud Native Computing Foundation (CNCF), KubeCon + CloudNativeCon North America 2023, Chicago, IL. *Awarded by CNCF on a competitive basis to support attendance and presentation at the world's largest cloud-native computing conference.*
- **Graduate Research Assistantship**, Rutgers University (WINLAB), competitive full-funding award covering tuition and stipend for the duration of PhD studies (2018–2024), granted on the basis of research merit and advisor recommendation.
- **Teaching Assistantship**, Rutgers University, Department of Electrical and Computer Engineering.
- **IEEE Member** — Institute of Electrical and Electronics Engineers, the world's largest technical professional organization.

Technical Skills

Methods: Queueing theory, Lyapunov optimization, stochastic processes, convex optimization, Age of Information analysis
Systems: Kubernetes, OpenShift, DPDK, NS-3, SUMO
Languages: Python, Go, C, C++, MATLAB, SQL, Verilog